ALTITUDE-TARGETING HIGH-POWER ROCKET SYSTEM

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Motivation

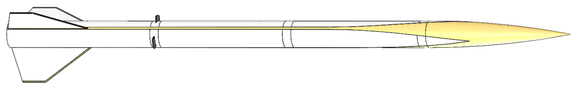
The George Washington University Rocket Team competes in the International Rocket Engineering Competition (IREC), where achieving a precise target altitude is a key criterion for evaluating team performance. This capstone project enables altitude targeting through the development of an airbrake system capable of in-flight correction. By integrating sensors, a controller, and an extruding brake mechanism, the system enables real-time adjustments to improve altitude accuracy leading to better performance in competition.

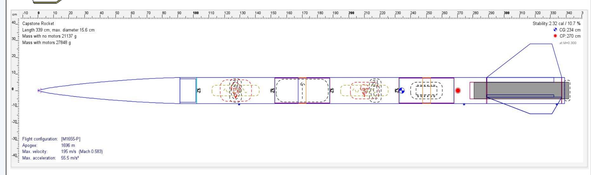
Objective

- Design and manufacture a high-power recoverable rocket, in the style of an IREC team rocket, to serve as a test vehicle.
- Design and integrate an airbrake system for in-flight altitude control, capable of reducing the rocket's apogee.
- Test-fly the rocket prior to airbrake system testing to ensure its function and the safety of the brakes.
- Evaluate the airbrake system's performance through flight tests.

Rocket Design

OpenRocket was used to design the rocket airframe, to evaluate its stability, and to aid engine sizing for a desirable apogee. The 11.12 ft tall rocket consists of fiberglass body tubes and two couplers; the aft coupler houses the airbrakes and the fore houses the recovery system. Components are held together with epoxy and bolts, and separating stages are secured during ascent with frangible nylon bolts. 4- to 6-gram black powder charges are used to separate stages and deploy parachutes.





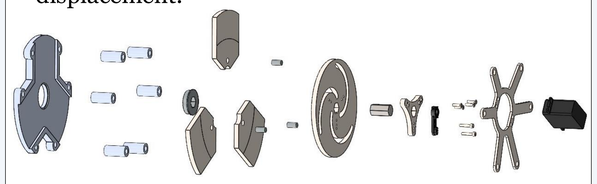
Propulsion System



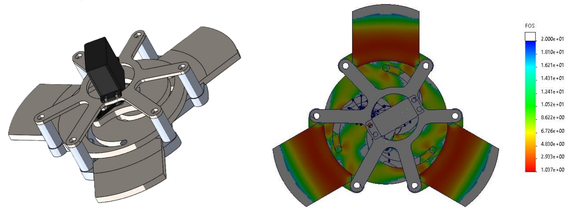
Custom propellant grains were manufactured to ensure flight testing closely reflected the GW Rocket Team approach for IREC, and were static fired to verify safe integration into the vehicle.

Mechanism Design

The airbrake mechanism is powered by a servomotor and driven by slot-pin linkages to reduce friction while preserving maximum linear displacement.



The mass, as dictated by the rocket's stability requirement, is reached via material selection. The system, including avionics and fixturing, weighs 12.125 lb.

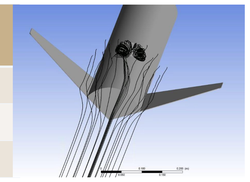


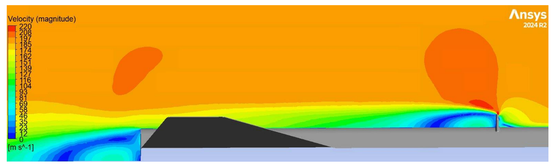
Finite-element analysis indicated that the mechanism will deform at a load of 72 times the expected drag, ensuring that a crash (and potentially an engine malfunction) will not cause damage.

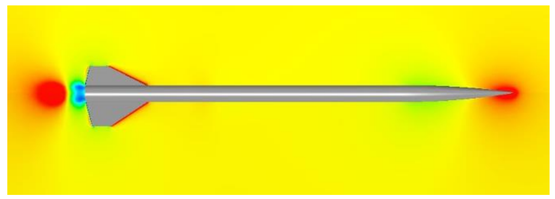
Computational Fluid Dynamics

CFD was used to evaluate the brakes' aerodynamic characteristics, in lieu of wind tunnel testing.

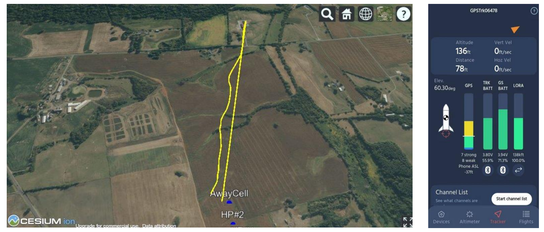
Brake Extrusion	Estimated Drag (N)
0% (clean)	201
50%	221.1
100%	270







Recovery Avionics

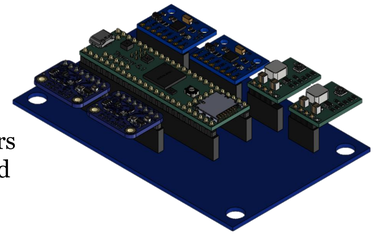


A dual-deployment architecture was implemented using an EasyMini as the primary system and an independent RRC3+ as a secondary to ensure reliable drogue and main parachute deployment. A Featherweight GPS provided real-time position tracking to support efficient recovery operations.

Airbrake Targeting Controller

The following components, sensors, and micro-controllers were used for data collection, computation, and control. The finalized wiring schematics were then used to design a PCB.

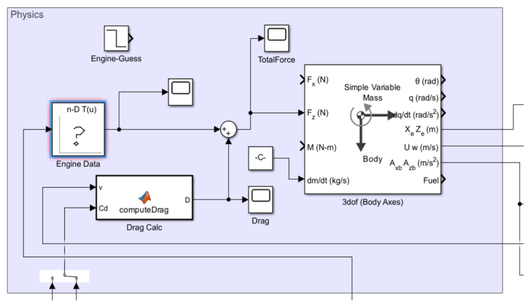
1x Teensy 4.1
2x BMP388
2x MPU6050
1x Servo
2x DC Converters
1x Micro SD card



The Airbrake followed 5 steps for control logic:

1. Preflight Setup
2. Launch and Burnout Detection
3. Apogee Prediction and Initial Fin Command
4. Feedback / Controller Loop
5. Fail-safes & Safety Logic

These steps were dependent on the state-estimation done through sensor acquisition and Kalman filtering.



A SIMULINK simulation was created to ensure the airbrake's capability of apogee reduction.

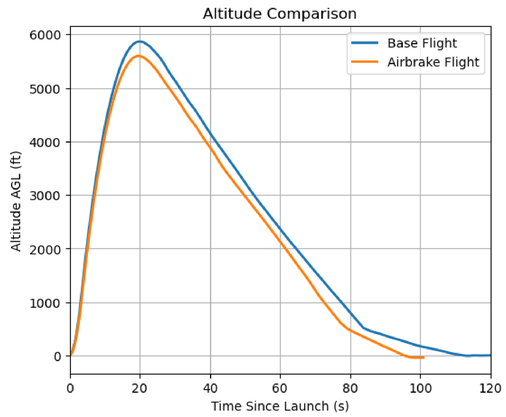
Flight Validation



A ground test was conducted to verify proper charge buildup for reliable stage separation. Flight performance was validated through three successful launch tests, supported by onboard camera footage.



Flight Results



The airbrake system successfully demonstrated functionality during launch, achieving a **4.6% reduction** in apogee from **5,867 ft** to **5,597 ft**.

Future Work

The GW Rocket Team now has validated airbrake designs and supporting test data, enabling integration as a selectable flight option for future IREC missions.

Acknowledgments

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